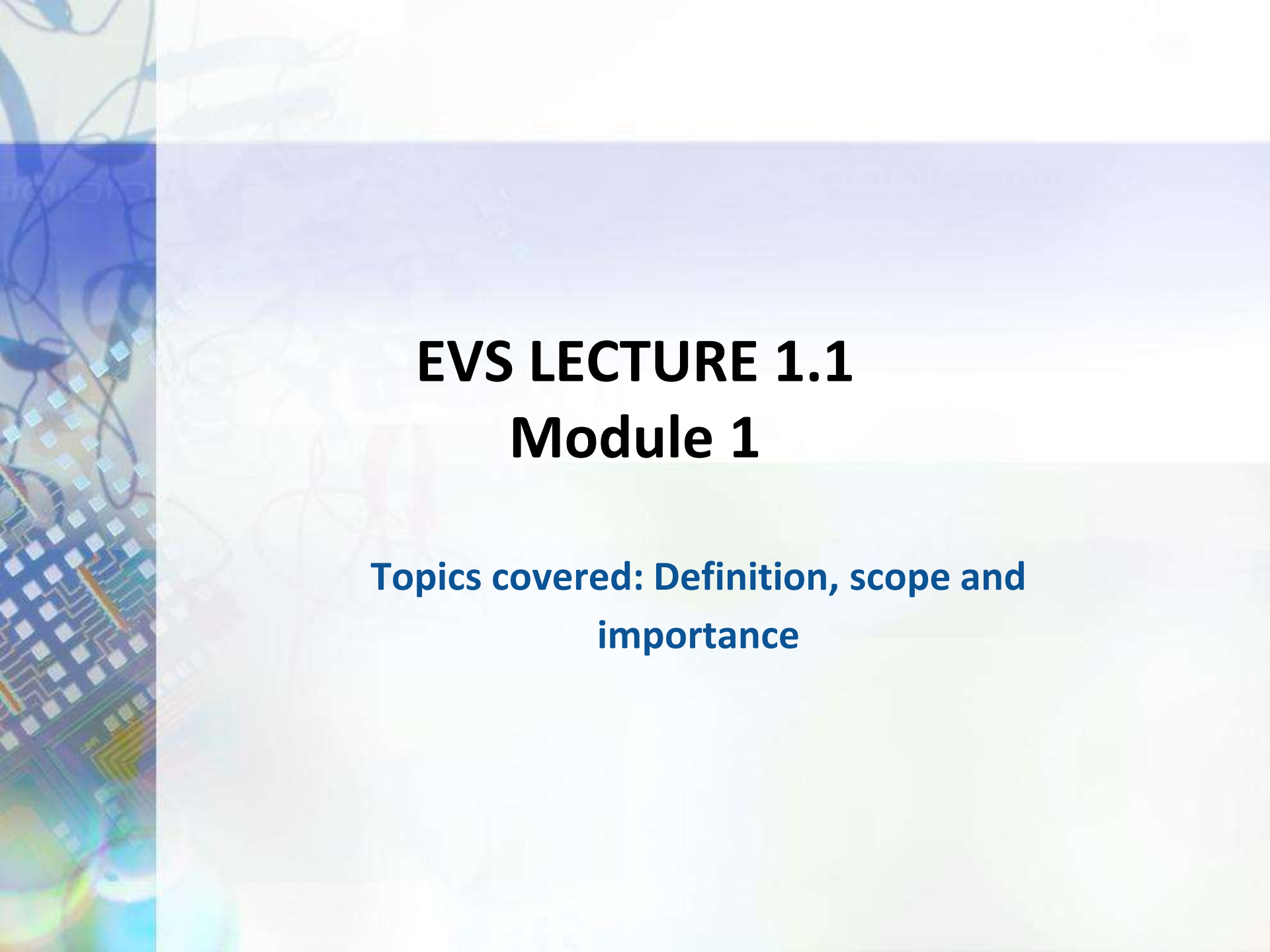


Environmental Studies

(19B13BT211)

Module 01: The Multidisciplinary nature of environment, Biodiversity

(Definition, scope and importance, Need for public awareness, Types of Ecosystems, World Biomes, Ecosystem functioning, Diversity of flora and fauna, species and wildlife diversity, Biodiversity hotspots, threats to biodiversity, Case studies)



EVS LECTURE 1.1

Module 1

Topics covered: Definition, scope and importance



Environmental studies

.....and what it is all about

Expected Course Outcome

COURSE OUTCOMES		COGNITIVE LEVELS
CO205.1	Explain diversity of environment, ecosystem resources and conservation .	Understand Level (C2)
CO205.2	Identify various pollution related hazard and their safe management	Apply Level (C3)
CO205.3	Apply modern techniques for sustainable Urban planning and Disaster management	Apply Level (C3)
CO205.4	Recall Government regulations, Environmental Policies, Laws & ethics	Understand Level (C2)
CO205.5	Survey ground situation on specific environmental aspects, examine risks involved, make a field report and present the findings	Analyzing Level (C4)

Reasons to study

- Clarifies modern environmental concepts
 - Equitable resources utilization
 - Sustainability
- To change the way we view environment.
- To create concern for environment.



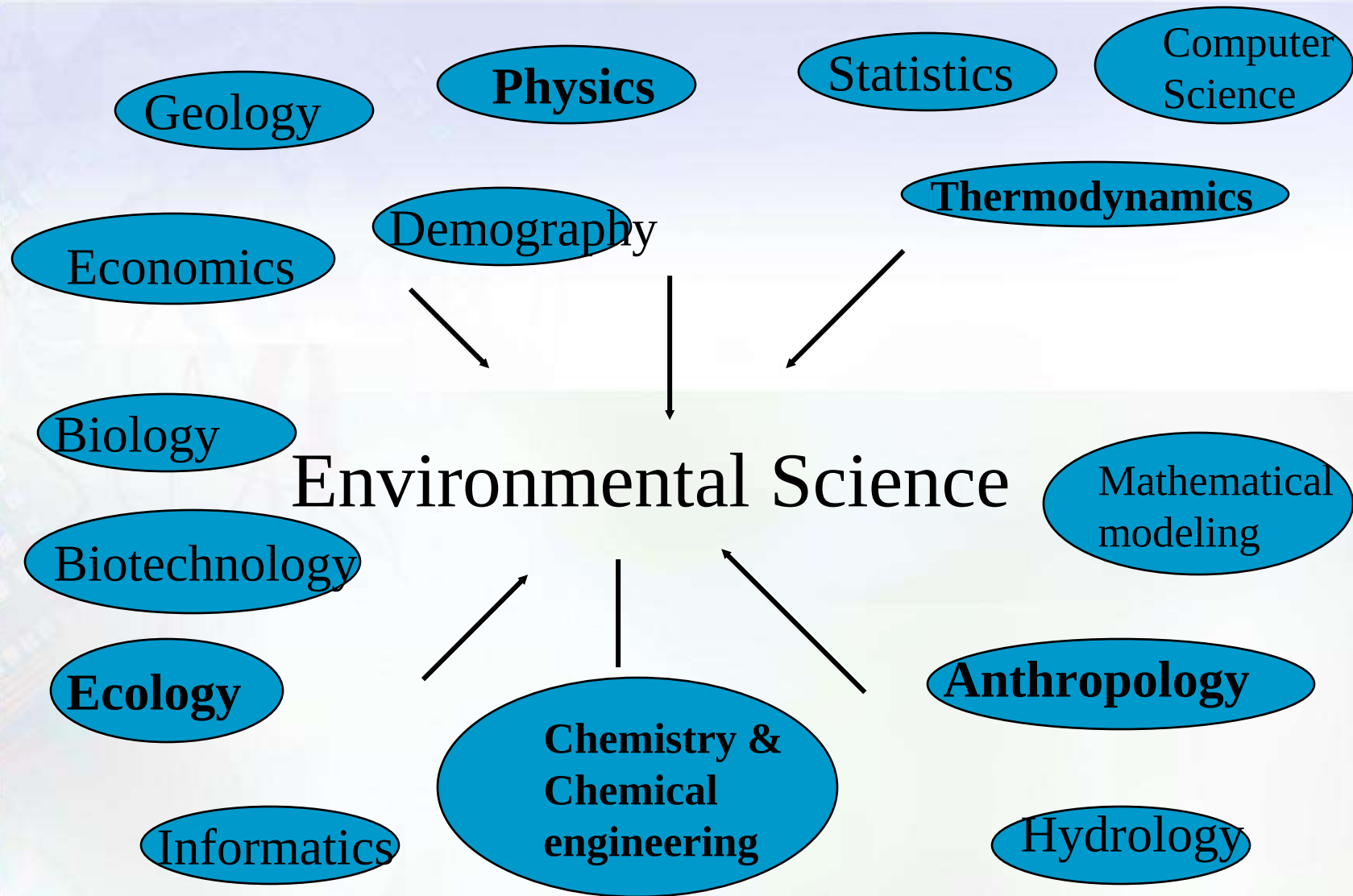
Environment

***The physical and biotic
habitat that surrounds us***

Environmental Science

- The study of the environment, its living and nonliving components, and the interactions of these components
- The discipline deals with the social, political, philosophical and ethical issues concerning man's interactions with the environment

The multidisciplinary integration of sciences and engineering in order to utilize the huge biochemical potential of microorganisms, plants and parts thereof for the restoration and preservation of the environment and for the sustainable use of resources



Why is the importance...?

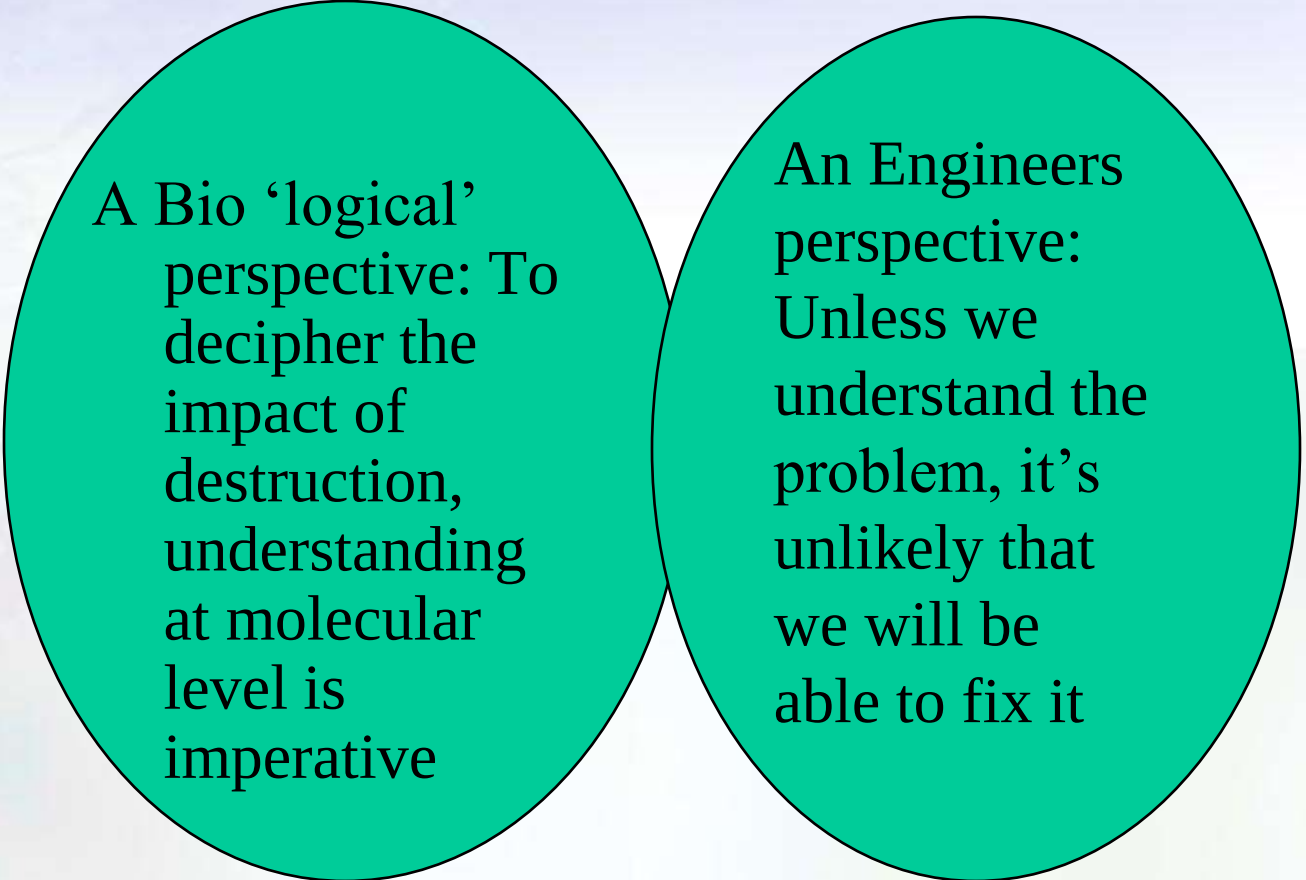
“The earth provides enough to satisfy every person's need but not greed”-MG

- The “**Laws of nature**” have not been repealed
- Man and his **power to alter** the nature's course have multiplied
- We the human race is only **one amongst millions** of different kinds of organisms that share the planet with

- 
- “Global village”
 - The Vedic declaration that all humanity is a single family -
vasudhaika kutumbam
 - Common environment
 - Transcending national boundaries
 - Growing concerns & Demands

Awareness is the Need of the hour....

- Environmental Literacy
 - Arise consciousness
 - Environmentally appropriate
 - Ecological interdependence of economic, social and political factors with environment
- Capacity building for
 - Analyze
 - Synthesis
 - Evaluate
 - Predict
 - Informed decision



A Bio 'logical' perspective: To decipher the impact of destruction, understanding at molecular level is imperative

An Engineers perspective: Unless we understand the problem, it's unlikely that we will be able to fix it

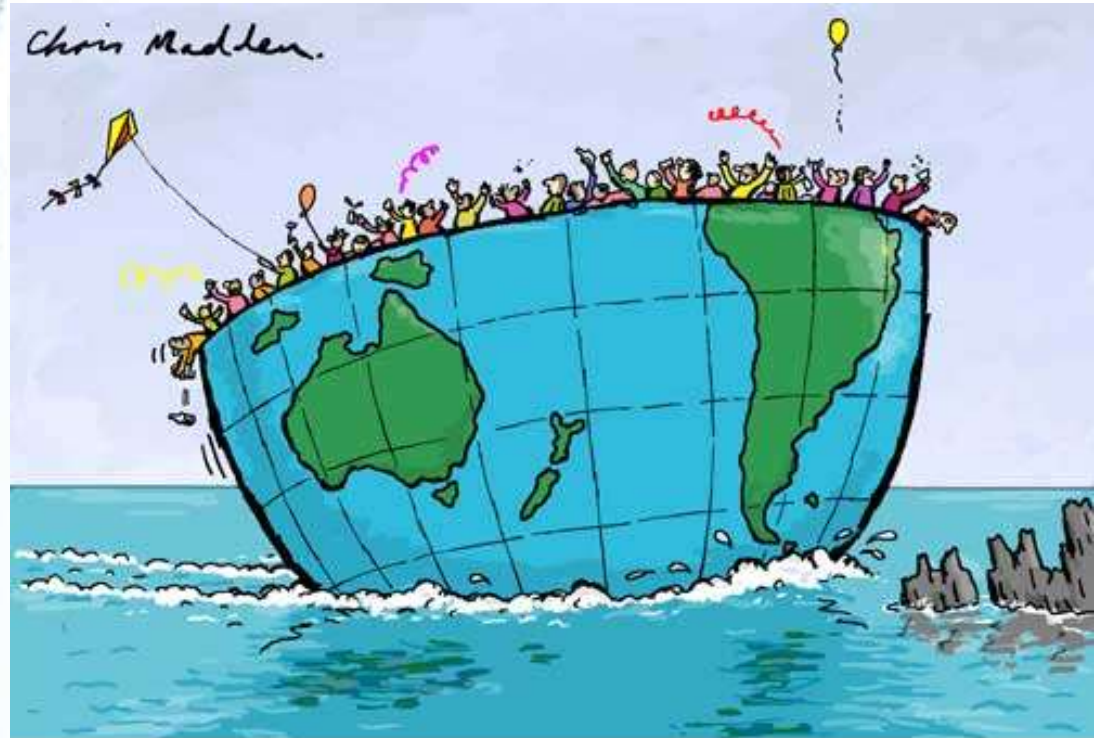
Rights vs Responsibilities

Conflict of development strategies with environmental protection

- Many technological advances stem from the scientific study of life and to devise tools for creating and maintaining human civilization
- Tools to create and maintain the quality of human civilization have become indispensable
- Man being part of nature, it is all the more important to study the structure and function of nature

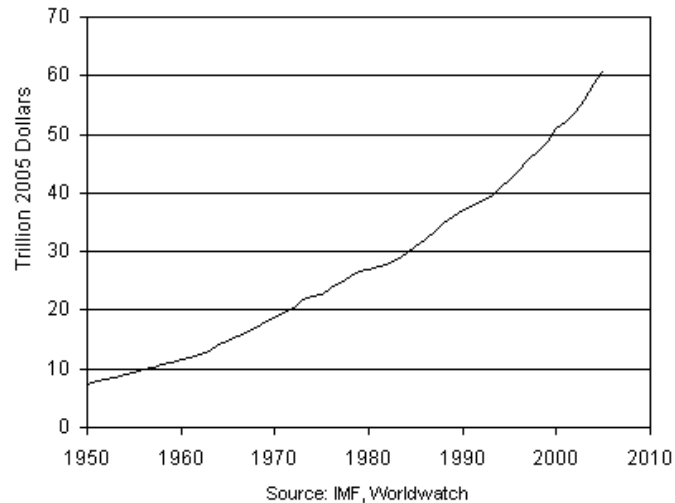


Environmental damage
increases as a country
becomes richer



Economic Growth vs Environment

Gross World Product, 1950-2005



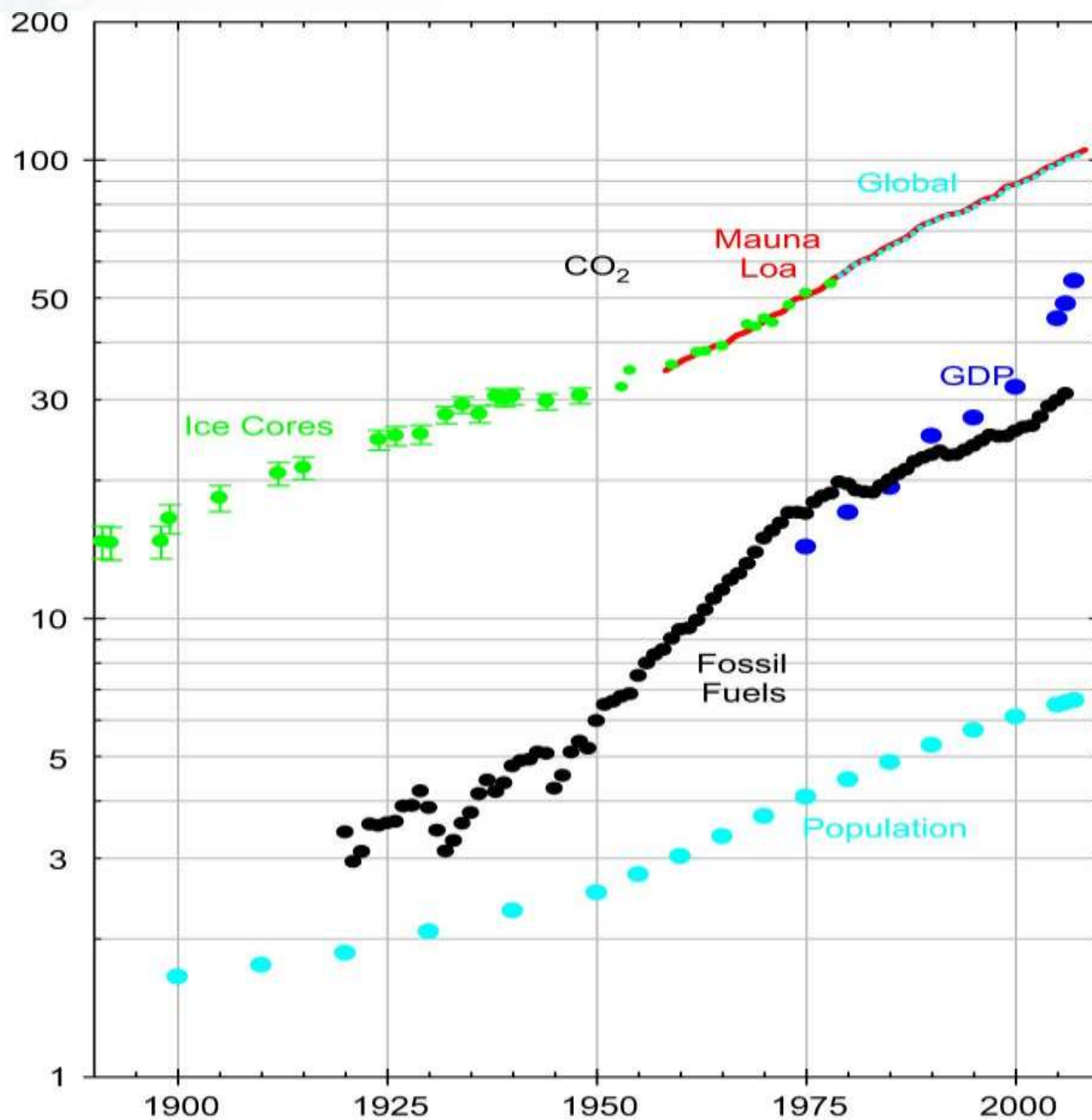
World GDP and CO2



Anthropogenic
CO₂ (ppm)

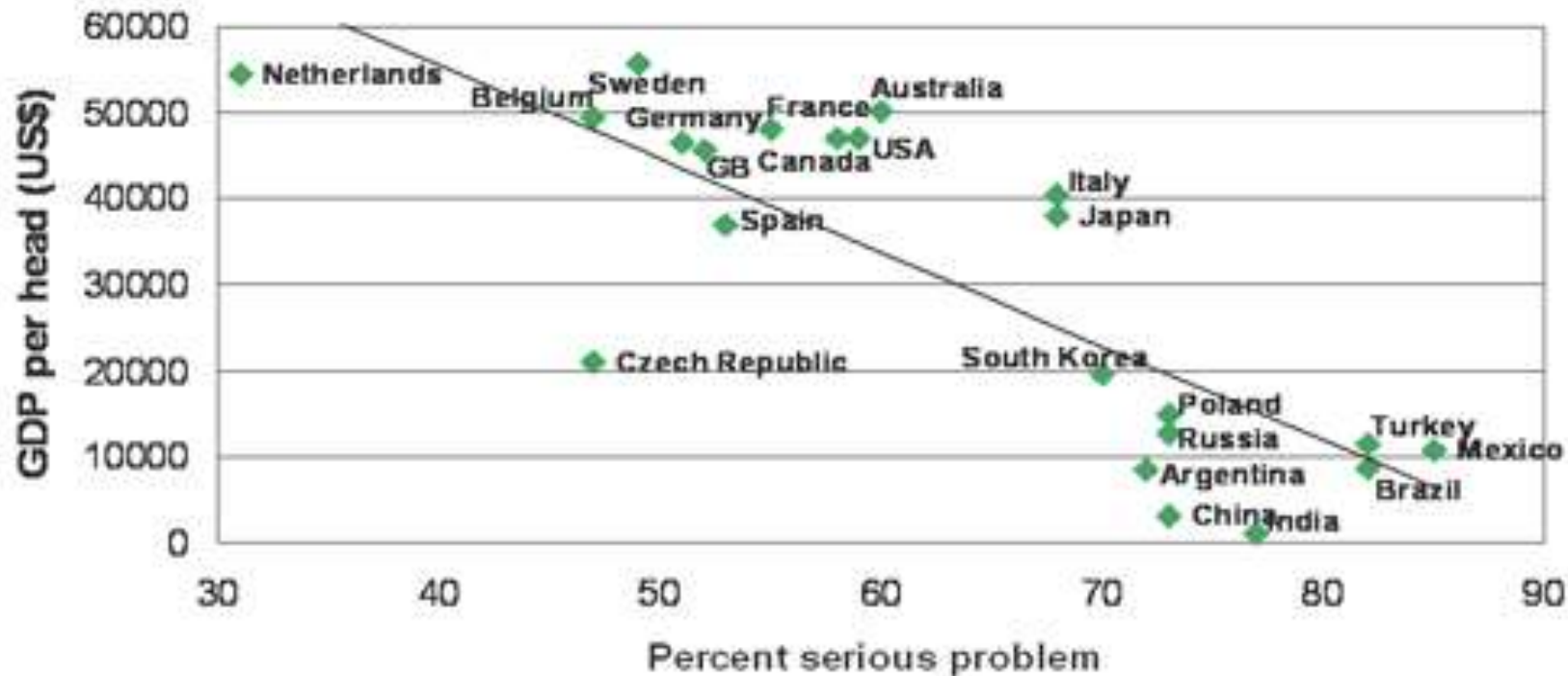
Fossil Fuel Emissions (billion tonnes CO₂ per year)

GDP (trillion US dollars), Population (billions)



As GDP rises, environmental concern falls

How serious a problem is the environment in ... ?



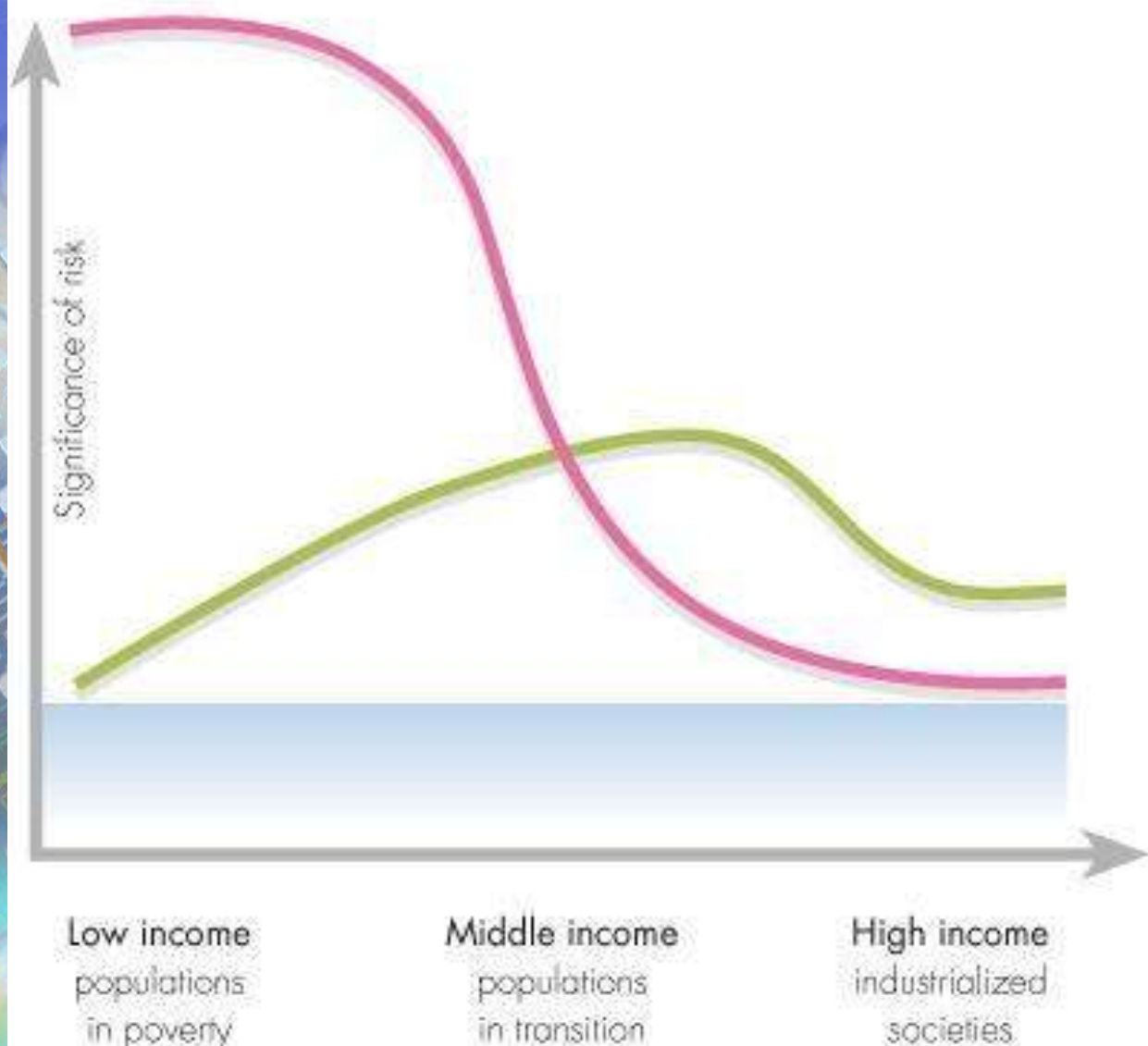
GDP from IMF 2008

Ipsos MORI

Base: 11,722 Online consumer citizens, November 2008

- Richer a nation – Weaker the sense of personal responsibility
- Emerging economies – Profess stronger sense of responsibility
 - «Minimising my impact on the environment is an important part of my life »

Environmental health risk transitions



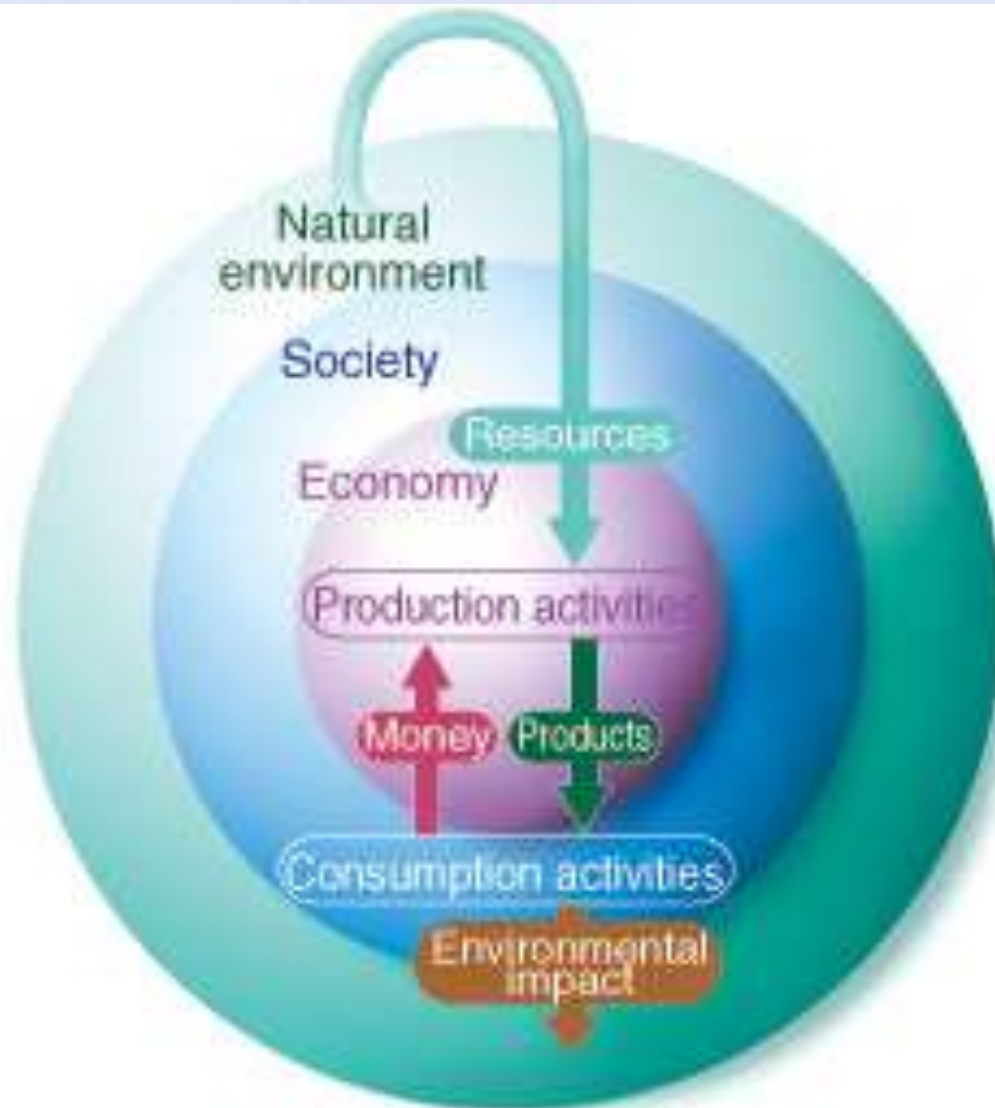
Summary of risks by income 2004

- Basic risks: lack of safe water, sanitation and hygiene, indoor air pollution, vector-borne diseases, hazards that cause accidents and injuries
- Modern risks: unsafe use of chemicals, environmental degradation
- Emerging risks: climate change, persistent organic pollutants, endocrine disruptors

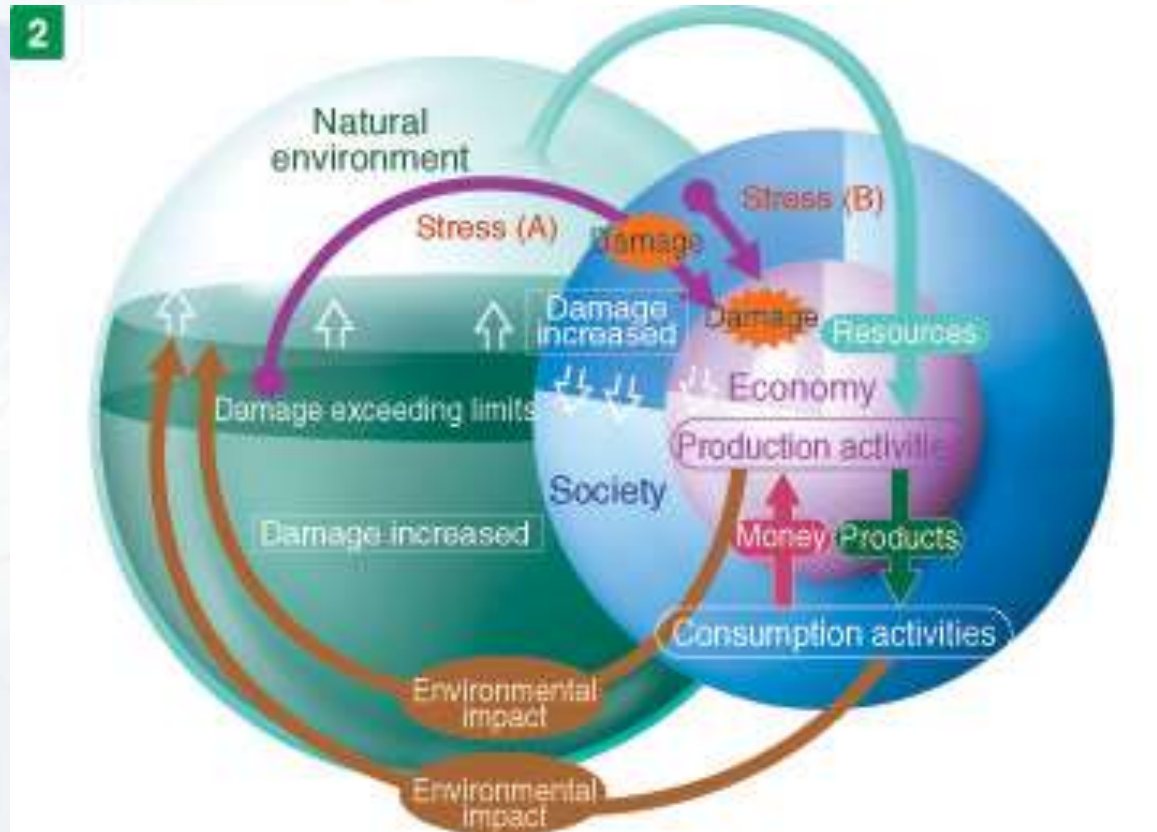
Source: Adapted from Gordon and others 2004

How is it before all this environmental deterioration started....?

1



Where and how did the problem arise....?



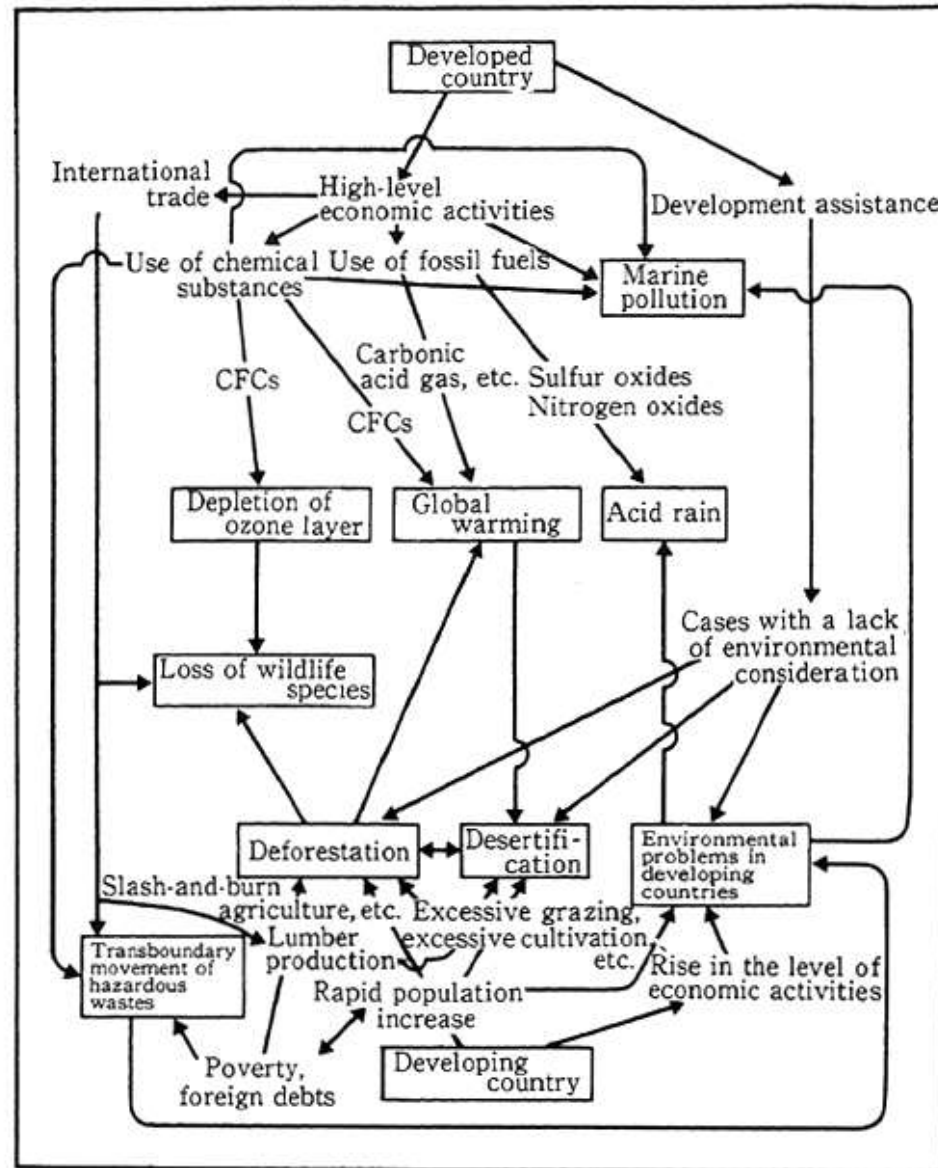
- Industrial revolution
- Green revolution

Leading to

- Mass production
- Mass consumption
- Mass disposal

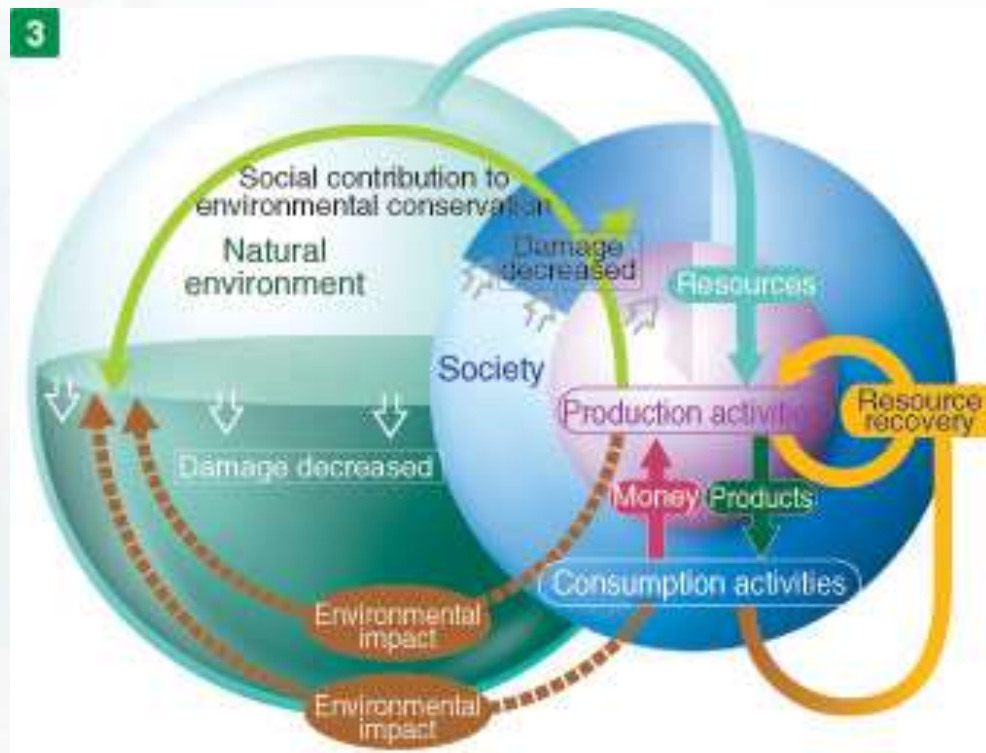
“Developing” or the “Developed”

Who is responsible?

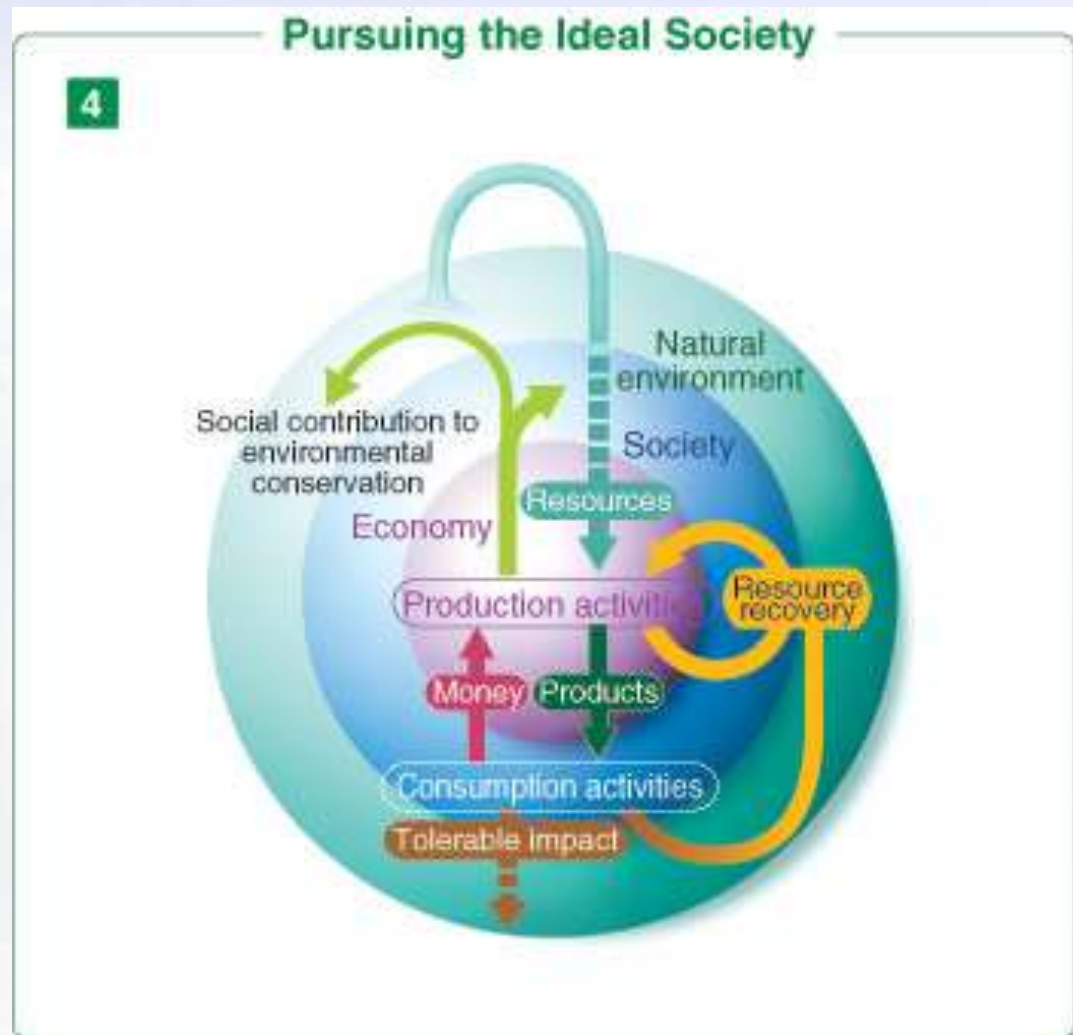


Course objective

Awaken Eco - 'logical' sense and trigger pro environmental approach in action, incorporating simple solutions in day to day developments



The ideal scenario





When we permit the destruction of
environment
and

Do not protest / raise our voice against it,
We are denying our future generations
the use of this valuable resource...
‘THE BIOSPHERE’